

Speech Recognition Spring 2025 Project 4

Due date: Feb 04th 2026 12pm Beijing time

In this assignment, you will implement and explore word embeddings by completing the provided Jupyter notebook `exploring_word_vectors.ipynb`. You will learn how to train a word2vec model and investigate the semantic relationships and potential biases encoded in these vector representations.

Tasks:

1. Word Vector Fundamentals & Visualization
 - Implement dimensionality reduction using Truncated SVD to project word vectors into 2D space.
 - Implement a function to visualize word embeddings in scatter plots.
2. Prediction-Based Word Vectors with Word2Vec
 - Implement a function to train a custom Word2Vec model on a text corpus. (you can use the training receipt in the dive into deep learning textbook.)
 - Load and explore pre-trained and custom-trained word vectors. (you can load a pre-trained model from the `gensim` library.)
 - Compute word similarities and solve analogy tasks using vector arithmetic.
 - Investigate words with multiple meanings and analyze semantic relationships.
 - Analysis social bias in word vectors.

Requirements:

Follow the detailed instructions in the notebook, complete all code sections, and answer the written questions to demonstrate your understanding of word vector methods and their applications.

Note:

This assignment is adapted from materials originally developed for Stanford's CS224N course.